Blindfold runs an AI model on any number of servers without any of them seeing your input.

Stays on your computer	Goes to each server
Your actual text	A random-looking piece of it
The final answer	The public model + random masks

Who can read your input

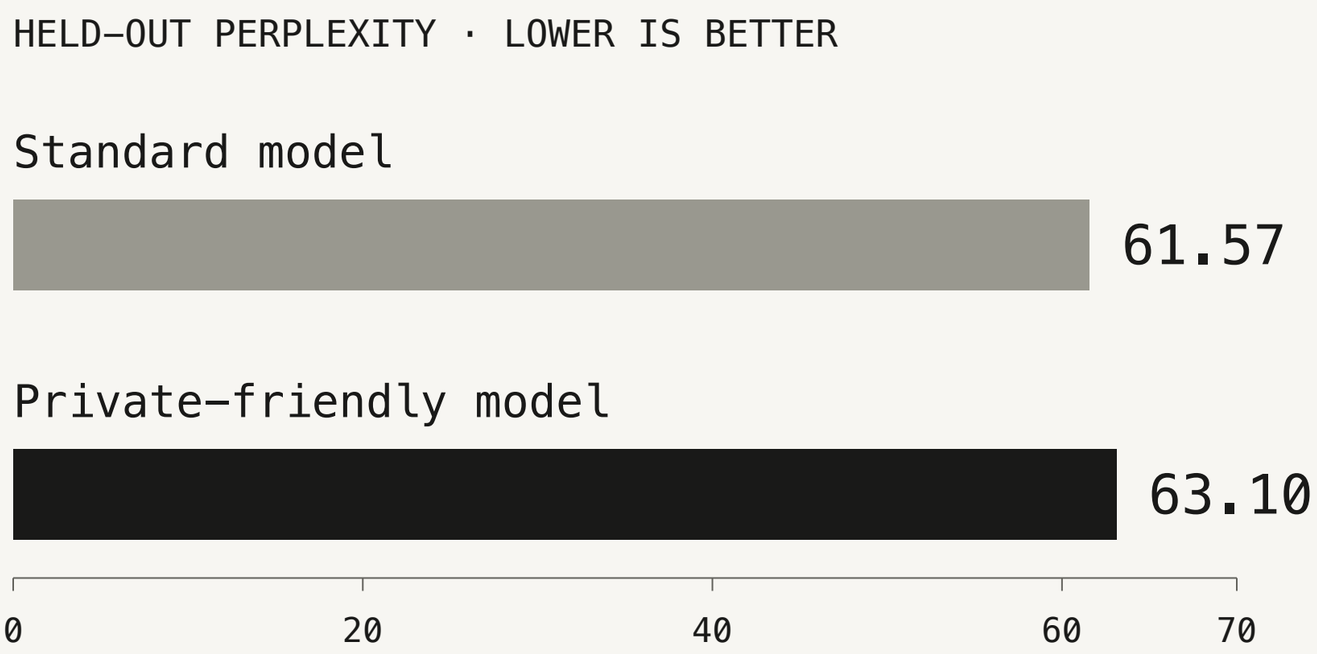
If these servers' pieces are combined	Can your input be read?
Any one server	No
All but one of the servers	No
All of the servers	Yes
One of them is yours	No

Anyone with every piece can read the input, and that can't be detected.

Assumes the servers are run by different people

In the demo, they all run on our desk

In a separate test, the private-friendly model was 2.48% worse



6,164,096
parameters in each test model

14,336,000
training tokens

Not the demo model.
About quality, not speed.

Measured: perplexity on held-out text · data/quality.json

Separate from the demo

The model only adds, multiplies and squares

Step	Who does it	Messages
Split the input into pieces	Your computer	Pieces go to the servers
Multiply by public weights	Each server, on its own	None needed
Square	Servers, through your computer	Checked before it’s used
Multiply again, then check	Servers, then your computer	Answer pieces come back
Per character		2 round trips

From engine/device.py and engine/node.py

Built vs. not built yet

Built	Not built yet
A 19,295-parameter test model	Real-size language models
Any number of servers (tested 2–7), over the network or USB	Servers run by different people
Same output as the normal model	A full security review
Your computer makes the random masks	Servers make them and talk directly